Xiaoqin Feng

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Curriculum Vitae

Los Angeles, CA 90017

Introduction

Master's student at the University of Southern California (USC) specializing in Applied AI, with a focus on LLM-based agent systems, generative models, and production-scale AI deployment. My research and engineering work includes building LLM-driven agentic systems, speech and NLP models, large-scale data processing pipelines, rigorous model evaluation frameworks, and reliable backend services.

Education

M.S. in Artificial Intelligence, University of Southern California (USC) — Sep. 2025 – May 2027 (expected)
M.S. in Software Engineering, Beijing University of Technology (BJUT) — Sep. 2016 – May 2019
B.E. in Computer Science, Southwest Minzu University (SMU) - "Top 5" of 154 students — Sep. 2012 – Jul. 2016

Professional Experience

Research Assistant [University of Southern California] **Oct. 2025 – Present**

- Conduct research on LLM-based agentic systems, including computer-use agents, research agents, and agent safety mechanisms, focusing on reliability, tool-use planning, and failure analysis.

Part-time Intern [Mobvoi - IT Company] **Aug. 2025 – Present**

- Optimized production speech models and LLM-based multi-agent workflows; supported data pipelines and system integration for reliable deployment.

Tech Lead [Mobvoi] **May. 2023 – Jul. 2025**

- **Agents:** Built a production-grade LLM agent framework (tool calling, multi-agent orchestration, safety guardrails, runtime monitoring, anomaly detection) for generative AI products, significantly improving creator productivity and user experience.
- **Speech-LLM:** Launched and iterated multiple generations of large-scale speech-LLM models enabling high-fidelity speech synthesis, 3-second voice cloning, dialect and style transfer, and fine-grained controllability; achieved end-to-end RTF of 0.3 and MOS scores of 4.8–4.9 in production.
- **Backend:** Designed and deployed distributed speech-LLM services for consumer AI products and API clients, sustaining 99%+ availability and processing millions of daily requests at scale.
- **Data Engineering:** Led large-scale multimodal (text/audio/image) data engineering, establishing data collection standards, dataset versioning, automated quality control, and scalable training/evaluation pipelines with strong emphasis on data safety, compliance, and responsible AI practices. Managed 4M+ hours of high-quality audio data, enabling data commercialization initiatives that generated seven-figure revenue.
- **Evaluation:** Built comprehensive evaluation systems for LLM, speech, and video models, covering algorithmic metrics and real-world product scenarios. Established human evaluation benchmarks and automated regression pipelines, reducing evaluation turnaround to 5–24 hours with automatically generated, production-ready reports. Integrated online user data into continuous feedback loops to enable data-driven optimization and stable model releases.
- **Collaboration:** Led cross-functional coordination across data, product, marketing, and engineering teams, aligning resources and driving end-to-end AI feature delivery from development to product launch.
- **Leadership:** Led and mentored 3+ junior engineers/interns, providing technical guidance across ML, data, and product delivery.

Speech Algorithm Engineer [Mobvoi] **Jul. 2019 – May. 2023**

- **Speech & NLP:** Led development and optimization of core speech and NLP modules, including multilingual TTS, text normalization, G2P conversion, prosody modeling, and emotion recognition, enhancing model accuracy, naturalness, and robustness in production systems.
- **Backend & Services:** Designed and built C++ speech backend services exposing scalable TTS/NLP APIs that powered a consumer AI product with tens of millions of users, improving large-scale service reliability to 99% availability through deployment hardening, monitoring, performance optimization, and incident response.
- **Data:** Established standardized annotation guidelines, workflows, and quality assurance processes for speech data representation, including prosody, pronunciation, emotion, and stress. Led the data team to produce 10,000+ hours of high-quality annotated audio across 8+ languages.
- **Collaboration:** Drove cross-functional delivery with PM and infra teams to ship AI features from prototype to production.

Research Intern [TAL Education Group] **Aug. 2018 – Dec. 2018**

- Developed a Deep Knowledge Tracing (DKT) model using graph-based embeddings and distance metrics to model and evaluate student knowledge progression.
- Co-authored a paper accepted at *AIED 2019*; received the Outstanding Intern Award for research contributions.

Projects

AI-Powered Audio Visualization Platform [https://qinzhi.org] **Nov. 2025 – Present**

- Designed and deployed an AI-driven audio visualization and dubbing platform, implementing scalable serverless backend infrastructure on GCP to enable high-fidelity speech processing and real-time interactive editing.

Computer Use Agent Research [GitHub] **Nov. 2025 – Feb. 2026**

- Conducted empirical research on Computer Use Agents (CUA) using **OS-World**, integrating world models to enhance autonomous decision-making, cross-interface generalization, and zero-shot performance in real-world computer interaction tasks.

AI For Education Research [GitHub] **Oct. 2025 – Dec. 2025**

- Collaborated on a research survey in AI-driven educational benchmarking, combining structured literature analysis with empirical validation. Applied LLM-based optimization techniques—prompt engineering, agent workflows, synthetic data augmentation, and supervised fine-tuning (SFT)—to improve model performance across educational benchmarks.

Technical Skills

- **Languages**: Python, C/C++, Bash/Shell, SQL, TypeScript
- **AI and Machine Learning**: LLM integration, agent systems, MCP, prompt engineering, benchmark evaluation, speech & NLP, multimodal learning.
- **Backend and Systems**: backend services, APIs, distributed systems, AI model serving, monitoring, CI/CD
- **Data Engineering**: large-scale datasets, data analysis, dataset construction, data quality, visualization and versioning.
- **DevOps and ML Tools**: Git, Docker, Linux, Kubernetes, cloud infrastructure(AWS, GCP, Azure); MCP, PyTorch, TensorFlow, scikit-learn, pandas, numpy, vLLM, Streamlit, ONNX, LoRA.
- **Vibe Coding**: Cursor, Antigravity, Claude Code, Windsurf.

Teaching Experience

Lecture - Embedded System Design Practice **2018**
Beijing University of Technology *Beijing, China*
Supported instruction and labs for $\tilde{8}0$ M.S. students annually.

Company - Speech & NLP **2021-2025**
Mobvoi *Beijing, China*
Mentored 3 interns on speech/NLP projects (annual program).

Publications

Proceedings

- Xinsheng W, **Xiaoqin F**, et al. *Spark-TTS: An Efficient LLM-Based Text-to-Speech Model with Single-Stream Decoupled Speech Tokens.* arXiv preprint, 2025. [PDF]
- Dongsheng W, **Xiaoqin F**, Zeming L, et al. *2M-NER: Contrastive Learning for Multilingual and Multimodal NER with Language and Modality Fusion.* Applied Intelligence, 2024. [PDF]
- Wenjiang C, **Xiaoqin F**(† equal contribution), Yunlin C, et al. *Multi-granularity Semantic and Acoustic Stress Prediction for Expressive TTS.* APSIPA ASC, 2023. [PDF]
- Jipeng Z, **Xiaoqin F**. *Prosody Prediction with Discriminative Representation Method.* PRML, 2022. [PDF]
- **Xiaoqin F**, Xie R, et al. *Population Statistics Algorithm Based on MobileNet.* Journal of Physics: Conference Series (JPCS), 2019. [PDF]
- Zhiwei W, **Xiaoqin F**, et al. *Deep Knowledge Tracing with Side Information.* AIED, 2019. [PDF]
- Xie Rong, **Xiaoqin F**. *A Method of Quick Edge Detection Based on Zynq.* CCIOT, 2018. [PDF]
- Sheng J, **Xiaoqin F**, et al. *Research on the Internet of Things Platform for Smart and Environmental Protection.* CCIS, 2018. [PDF]

Patents

- all relevant to speech/NLP/LLM areas.
  6 First inventor: CN111078898A CN110970013A CN111145724A CN115470351A CN115470350A CN115547289A
- **4 Co-inventor**: CN111079428A CN111178042A CN115578998A CN116013251A